

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence (AI) and Data Science) SEMESTER - 3 Winter 2025 (Regular)

Course : Bachelor of Technology (Artificial Intelligence (AI) and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: 25AFAIMD306A - ARTIFICIAL INTELLIGENCE

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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- 1 Artificial Intelligence (AI) is primarily concerned with:
A. Studying human anatomy
B. Creating machines that can think and act intelligently
C. Developing faster hardware
D. Improving database storage
- 2 Which of the following best defines an intelligent agent?
A. A machine that stores data
B. An entity that perceives its environment and acts upon it
C. A software program without input
D. A human-like robot only
- 3 The concept of "rationality" in AI refers to:
A. Acting emotionally
B. Acting randomly
C. Doing the right thing based on available information
D. Acting only on human commands
- 4 Which search strategy explores the search tree level by level?
A. Depth First Search
B. Hill Climbing
C. Breadth First Search
D. A* Search
- 5 In problem-solving, the initial state represents:
A. Final solution
B. Starting point of the problem
C. Cost function
D. Goal test
- 6 Which logic is mainly used for knowledge representation in AI?
A. Boolean logic
B. Predicate logic
C. Fuzzy logic
D. Temporal logic
- 7 Bayesian Networks are mainly used for:
A. Game playing
B. Logical reasoning
C. Reasoning under uncertainty
D. Pattern matching
- 8 Fuzzy logic deals with:
A. Only true or false values
B. Random events
C. Degrees of truth
D. Binary variables
- 9 Which uninformed search strategy may get trapped in infinite depth?
A. Breadth First Search
B. Uniform Cost Search
C. Depth First Search
D. A* Search

- 10 Forward reasoning in rule-based systems is also known as:
 A. Goal-driven reasoning B. Backward chaining
 C. Data-driven reasoning D. Constraint propagation
- 11 Constraint Satisfaction Problems (CSPs) are defined by:
 A. States, actions, and rewards B. Variables, domains, and constraints
 C. Nodes and edges D. Rules and facts
- 12 Alpha-Beta pruning improves minimax search by:
 A. Expanding all nodes B. Increasing branching factor
 C. Eliminating branches that cannot affect the final decision
 D. Randomly pruning nodes

Q2. Solve the following.

A) Explain the term nature of environments in AI.

B) Describe the structure of an intelligent agent.

Q3. Solve the following.

A) Explain different types of agents with a suitable diagram.

B) Describe how problem-solving performance is measured.

Q4. Solve Any Two of the following.

A) Explain the working of the Breadth-First Search (BFS) algorithm by describing the following

a) Working principle b) Algorithmic steps

c) Illustrative example d) Advantages and disadvantages

B) Demonstrate the use of Heuristic Functions in solving a search problem with a suitable example.

C) Differentiate between procedural knowledge and declarative knowledge.

Q5. Solve Any Two of the following.

A) Illustrate how knowledge is represented in AI systems using a suitable diagram.

B) Apply various knowledge representation techniques to represent a given problem domain.

C) Derive the resolution principle in predicate logic to derive a conclusion from a given set of predicates, illustrating the process with a suitable example.

Q6. Solve Any Two of the following.

A) What are Bayesian networks? Explain their semantics with an example.

B) What is probabilistic reasoning? Explain how knowledge can be represented in an uncertain domain.

C) What are Fuzzy sets and the various operations that can be performed with Fuzzy sets?

*** End ***